

зг. 1.1) $H \rightarrow 1 \text{ зап}$
 $3 \rightarrow 2 \text{ зап}$
 $C \rightarrow 3 \text{ зап}$

$$P(\text{3-я линия 3}) = \frac{P(\text{3-я линия 3} | \text{3-я линия 3}) P(\text{3-я линия 3})}{P(\text{3-я линия 3})}$$

$$= \frac{\frac{2}{36} \cdot \frac{1}{3}}{\frac{1}{6} \cdot \frac{1}{6} + \frac{2}{36} \cdot \frac{1}{3} + \frac{1}{216} \cdot \frac{1}{2}} = \frac{8}{21} \approx 38.10\%$$

$$2) P(\max \geq 3) = 1 - P(\max = 1 \text{ или } 2) \\ = 1 - \left(\frac{1}{3}\right)^5$$

$P(10 / \text{произведено}) = ?$

Нека $A := \{ \text{шар черно, което се глуми на 5} \}$
 $B := \{ \text{шар бял, който се глуми на 2} \}$

$$P(A) = \left(\frac{5}{6}\right)^5, P(B) = \left(\frac{1}{2}\right)^5; P(A \cap B) = \left(\frac{2}{6}\right)^5$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 1 - P(\bar{A}) - (1 - P(B)) - (1 - P(A \cap B))$$

$$= 1 - \left(\frac{5}{6}\right)^5 - \left(\frac{1}{2}\right)^5 + \left(\frac{1}{6}\right)^5 \approx 56.70\%$$

$$3) P(X_3 = 2) = P(2 \text{ кутии с 3 точки и 1 с точки})$$

$$= \frac{7 \cdot \binom{7}{3} 6 \binom{4}{3} \cdot 5/2}{7^7} \approx 1,78\%$$

$$\begin{aligned} P(X_3=1) &= P(1 \text{ куча с 3 тонна}) \\ &= P(3+4; 3+4 \times 1; 3+2 \times 2; \overset{3+2 \times 1}{7}) \\ &= \frac{7 \cdot \binom{7}{3} \cdot 6}{7^7} + \frac{7 \cdot \binom{7}{3} \cdot 6 \cdot 5 \cdot 4 \cdot 3}{7^7} \\ &\quad + \frac{7 \cdot \binom{7}{3} \cdot 6 \cdot \binom{4}{2} \cdot 5/2}{7^7} + \frac{7 \cdot \binom{7}{3} \cdot 6 \cdot \binom{4}{2} \cdot 5 \cdot 4}{7^7} \\ &\approx 34,99\% \end{aligned}$$

$$P(X_3=0) = 1 - P(X_3=0 \text{ или } 1) \approx 63,23\%$$

* Проверка:

$$\begin{aligned} P(X_3=0) &= P(7 \times 1 \text{ тонна}, 5 \times 1 \text{ тонна} + 1 \times 2 \text{ тонна}, \\ &\quad 3 \times 1 + 1 \times 4, 3 \times 1 + 2 \times 2, \\ &\quad 2 \times 1 + 1 \times 5, 1 \times 1 + 1 \times 2 + 1 \times 4, 1 \times 1 + 1 \times 6, \\ &\quad 1 \times 1 + 3 \times 2 \text{ тонна}, 0 \times 1 + 1 \times 2 + 4 \times 5) \end{aligned}$$

$$= \frac{7!}{7^7} + \frac{7 \cdot \binom{7}{2} \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{7^7} + \frac{7 \cdot \binom{7}{4} \cdot 6 \cdot 5 \cdot 4}{7^7}$$

$7 \times 1 \text{ тонна}$
 $\approx 0,61\%$

$1 \times 2 \text{ тонна} + 5 \times 1 \text{ тонна}$
 $\approx 12,85\%$

$1 \times 4 \text{ тонна} + 3 \times 1 \text{ тонна}$
 $\approx 3,57\%$

$$+ 7 \cdot \binom{7}{2} \cdot 6 \binom{5}{2} \cdot 5 \cdot 4 \cdot 3 / 2$$

$$+ \frac{7 \cdot \binom{7}{5} \cdot 6 \cdot 5}{7^7} + \frac{7 \cdot \binom{7}{4} \cdot 6 \cdot \binom{3}{2} \cdot 5}{7^7}$$

$$+ \frac{\binom{7}{3} \cdot \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{2!} \cdot 4}{7!} + \frac{7 \cdot \binom{7}{6} \cdot 6}{7^7} + \frac{7 \cdot \binom{7}{5} \cdot 6}{7^7}$$

$1 \times 1 \text{ τόνος} + 3 \times 2 \text{ τόνους} \approx 10.71\%$
 $1 \times 6 \text{ τόνους} + 1 \times 1 \text{ τόνος} \approx 0\%$
 $1 \times 5 \text{ τόνους} + 1 \times 2 \text{ τόνους} \approx 0\%$
 $\approx 63.3\%$

$$E[X_3] = 1 \cdot P(X_3=1) + 2 \cdot P(X_3=2)$$

$$\approx \frac{35}{100} + \frac{4}{100} \approx \frac{37}{100}$$

$$\sum_{i=0}^7 i \cdot E[X_i] = E[X_0 + 1 \cdot X_1 + \dots + 7 \cdot X_7] = 7, \text{ τζέ}$$

και $X_0 + 1 \cdot X_1 + 2 \cdot X_2 + \dots + 7 \cdot X_7 = \text{συνολικοί τόνοι}$

Αρα έχουμε $\sum_{i=0}^7 E[X_i] = E[X_0 + \dots + X_7] = E[\text{συνολικοί τόνοι}] = 7$

4) $E[XY] = E[X] E[Y] \vee g \rightarrow B$
 Βέβαια αληθές

$E[X/Y] \neq E[X] / E[Y]$ μόνο για
 ce συστήματα που τριπλασιάζουν το ποσό
 בכל המקרים X και Y . Δεν είναι

$$X \mid 1 \mid 2$$

$$\mid \frac{1}{2} \mid \frac{1}{2}$$

$$Y \mid 2 \mid 3$$

$$\mid \frac{1}{2} \mid \frac{1}{2}$$

заг. 2) $X = \# \text{ 6-ya} \sim \text{Bm}(6, \frac{1}{6})$

$Y = \text{позитивна}$ се зема при 1 хвърляне

$$P(Y=6) = \sum_{k=0}^6 P(Y=6 | X=k) P(X=k)$$

$$= \sum_{k=0}^6 \frac{k}{6} \cdot \binom{6}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{6-k}$$

$$= \frac{1}{6} E[\text{Bm}(6, \frac{1}{6})] = \frac{1}{6}.$$

$Z = \# \text{ хвърляния до } 1\text{-ва } 6\text{-за.}$

Взможно е да получим зора с

0 6-ya и това $Z | X=0 = \infty$

$\Rightarrow E[Z] = \infty.$ (т.к. никога няма да хвърлим 6)

2.) Аналогично, очакването е ∞ , т.к.

можем да получим зора само с 5-ya

3.) По Фелд: $P(\text{само 6-ya / надпопава})$

$$1 \cdot 1 \cdot P(X=6)$$

$$= \sum_{k=0}^6 P(X=k) \cdot P(\text{надпопава } 6 \text{ с } 6 | k)$$

$$= \frac{\left(\frac{1}{6}\right)^6}{\sum_{k=0}^6 \binom{6}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{6-k} \left(\frac{k}{6}\right)^2} = \frac{\left(\frac{1}{6}\right)^6}{\left(\frac{1}{6}\right)^2 \mathbb{E}[\text{Bin}(6, \frac{1}{6})^2]}$$

$$= \left(\frac{1}{6}\right)^4 \cdot \frac{1}{6 \cdot \frac{1}{6} \cdot \frac{5}{6} + \left(6 \cdot \frac{1}{6}\right)^2} \approx 0.04\%$$

За най-вероятното, трябва да сравним

$$f(k) = \frac{P(X=k) \cdot \left(\frac{k}{6}\right)^2}{\binom{6}{k} 5^{6-k} \cdot \left(\frac{k}{6}\right)^2} \quad \text{за } k=0, \dots, 6$$

k	0	1	2	3	4	5	6
f(k)	= 0	$\approx \frac{5}{83}$	$\approx \frac{10}{101.67}$	625	$\approx \frac{16}{67}$	$\approx \frac{20}{83}$	= 1

$\Rightarrow$ Най-вероятните записвания са тези с 2 б-ци.

Зад. 3 | Нека $\varphi_n(k) = P(\text{да сме избрали } k, \text{ записвайки с избор от } [1, n])$
 $= P(k \in E_n)$

Класово кодирование e, z

$$\phi_n(k) = \underbrace{\phi_n(k+1)}_{\substack{\text{ga uzberem} \\ k, \text{ bez ga uzberem} \\ k+1 (\text{zачо?})}} + \underbrace{\phi_n(k+1) \cdot \frac{1}{k}}_{\substack{\text{ga uzberem} \\ k+1 \text{ u sloz roba} \\ k}}$$

$$\Rightarrow \phi_n(k) = \frac{k+1}{k} \phi_n(k+1)$$

$$\text{Тзв казо } \phi_n(n) = \frac{1}{n} \text{ (или } \phi_n(1) = 1)$$

$$\text{Ито } \phi_n(k) = \frac{k+1}{k} \cdot \phi_n(k+1) = \frac{k+1}{k} \cdot \frac{k+2}{k+1} \phi_n(k+2)$$

$$= \dots = \frac{\cancel{k+1}}{k} \cdot \frac{\cancel{k+2}}{\cancel{k+1}} \dots \frac{\cancel{n}}{\cancel{n-1}} \phi_n(n)$$

$$= \frac{1}{k}$$

but no rope

$$2. P(2 \in E_n | 3 \notin E_n) = \frac{\phi_n(3)}{1 - \phi_n(3)} = \frac{1/3}{2/3}$$

$$= 50\%$$

$$3. E[|E \cup V|] = E\left[\sum_{i=1}^n \mathbb{1}_{\{i \in \text{uzbrano}\}}\right]$$

$$= \sum_{i=1}^n P(i \in \text{uzbrano})$$

$$= 1 + \dots + \frac{1}{n} \sim \ln(n) \quad \text{при } n \rightarrow \infty.$$

$$D(E_N) = \sum_{i,j} \text{Cov}(\underbrace{1_{\{X_i \in \text{уздырама}\}}}_{X_i}, 1_{\{X_j \in \text{уздырама}\}})$$

$$D(1_{\{X_i \in \text{уздырама}\}}) = \frac{1}{i} - \frac{1}{i^2}$$

$$\begin{aligned} \text{За } i < j, \quad \text{Cov}(X_i, X_j) &= E[X_i X_j] - E[X_i] E[X_j] \\ &= E[1_{\{\text{уздырама } i \text{ и } j \text{ одновременно}\}}] - \frac{1}{i} \cdot \frac{1}{j} \end{aligned}$$

$$\begin{aligned} P(\text{уздырама } i \text{ и } j \text{ одновременно}) &= P(A_{ij} | \text{уздырама } j) \cdot P(X_j = 1) \\ &= \frac{1}{i} \cdot \frac{1}{j} \text{ (защо?)} \end{aligned}$$

$$\Rightarrow \text{Cov}(X_i, X_j) = 0 \text{ за } i \neq j!$$

$$\begin{aligned} \Rightarrow D(|E_N|) &= D(\sum X_i) = \sum D X_i \\ &= 1 + \dots + \frac{1}{n} - \left[1 + \left(\frac{1}{2}\right)^2 + \dots + \frac{1}{n^2} \right] \\ &\sim \ln(n) \end{aligned}$$

$$1. \text{т.к. } \sum_{n \in \mathbb{N}} \frac{1}{n^2} < \infty$$

300g. 4

$X \backslash Z$	0	1	2	
0	$\frac{1}{9}$	0	$\frac{2}{9}$	$\frac{1}{3}$
1	$\frac{2}{9}$	$\frac{1}{9}$	0	$\frac{1}{3}$
2	0	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{1}{3}$
	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	

$$E[X] = E[Z] = 1$$

$$D X = D Z = \frac{1}{3} \cdot 1^2 + \frac{1}{3} 2^2 - 1$$

$$= \frac{2}{3}$$

$$\text{Cor}(X, Z) = \frac{E[XZ] - E[X] \cdot E[Z]}{\sqrt{D X \cdot D Z}}$$

$$= \frac{1 \cdot 1 \cdot \frac{1}{9} + 2 \cdot 1 \cdot \frac{2}{9} + 2 \cdot 2 \cdot \frac{1}{9} - 1^2}{\sqrt{\frac{4}{9}}}$$

$$= 0$$

$$2. \quad g_X(s) = \frac{1}{6} [s + \dots + s^6]$$

$$E[X] = g'_X(1) = 1$$

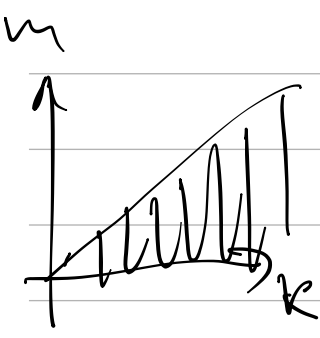
$$D X = g''_X(1) + g'_X(1) - (g'_X(1))^2$$

3. Ако $P(X=k, Y=m) = P(X=k)P(Y=m)$

$$ES^{X+Y} = \sum_{k=0}^{\infty} s^k P(X+Y=k)$$

$$= \sum_{k=0}^{\infty} s^k \sum_{m=0}^k P(X=m, Y=k-m)$$

т.е. $\sum_{k=0}^{\infty} s^k \sum_{m=0}^k P(X=m)P(Y=k-m)$



$$= \sum_{k=0}^{\infty} \sum_{m=0}^k s^m P(X=m) s^{k-m} P(Y=k-m)$$

$$= \sum_{m=0}^{\infty} s^m P(X=m) \sum_{k=m}^{\infty} s^{k-m} P(Y=k-m)$$

$$= g_X(s) g_Y(s).$$

Обратното не е вярно!

Контрпример е разпределението от
Таблица 1!

$$g_X(s) = g_Z(s) = (1+s+s^2)^2 \left(\frac{1}{3}\right)^2$$

$$= (s^4 + 2s^3 + 3s^2 + 2s + 1) \cdot \frac{1}{9}$$

$X+Z$	0	1	2	3	4
	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{2}{9}$	$\frac{1}{9}$

$$\Rightarrow g_{X+Z}(s) = g_X(s)g_Z(s)$$

~~$X \neq Z$~~ , $\neq k$.

$$P(X=0, Z=2) = 0 \neq P(X=0) \cdot P(Z=2)$$

4) Да, възможно е.

За стандартен зар $g_X(s) = (s + \dots + s^6) \frac{1}{6}$

$$\Rightarrow g_{X+X}(s) = (s + \dots + s^6)^2 \cdot \frac{1}{36}$$

Искаме да представим горното като произведение на 2 поранджени ф-и, различни от $\frac{1}{6}(s + \dots + s^6)$.

Това е еквивалентно да разложим

$$(s + \dots + s^6)^2 = P(s)Q(s), \text{ където}$$

$P(s)Q(s)$ са полиноми, $P(1) = Q(1) = 6$
и $P(0) = Q(0) = 1$ (защо?)

$$(s + \dots + s^6)^2 = [s(s^2 + s + 1)(s^3 + 1)]^2$$

$$= [s(s+1)(s^2 + s + 1)(s^2 - s + 1)]^2$$

$\begin{matrix} \text{ГТТ 6} & & & \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 1 & 2 & 3 & 1 \end{matrix}$

$$\Rightarrow \tau, k. \quad P(0) = Q(0) = 0 \quad P(1) = Q(1) = 6, \tau_0$$

$$P(s) = s(s+1)(s^2 + s + 1)$$

$$Q(s) = s(s+1)(s^2 + s + 1)$$

1 шаг) $P(s) = s(s+1)(s^2 + s + 1)$

$$Q(s) = P(s)(s^2 - s + 1)^2$$

$$\Rightarrow P(s) = s^4 + 2s^3 + 2s^2 + s$$

$$Q(s) = s^8 + s^6 + s^5 + s^4 + s^3 + s$$

$$\Rightarrow \text{запове с } 1, 2, 2, 3, 3, 4 \quad 4$$

$$1, 3, 4, 5, 6, 8 \quad \text{с} \quad 9$$

Решение та же.

2 шаг) $P(s) = Q(s) = s(s+1)(s^2 + s + 1)(s^2 - s + 1)$

клето бегу го 2 шаг